The Integer Traceability Chain

From SU(5) embedding to cosmological predictions in seven exact links.

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Abstract

The beta unification program (PHYS-25) found that the integers 13, 19, 20, and 22 — extracted from the gauge coupling beta coefficients — predict seven cosmological observables at sub-percent precision. This paper derives each integer from the gauge group alone, with no measurement, through a chain of seven exact links. The chain starts at the SU(5) embedding condition $\mathrm{Tr}(Y^2) = k_1\,\mathrm{Tr}(T_3^2)$, which gives the GUT normalization factor $k_1 = 3/5$ from explicit traces over one SM generation. From k_1, the Dynkin index coefficients (2/5, 2/3, 1/3) for vector-like fermion pairs follow. Applied to the Cabibbo Doublet (3,2,1/6), these produce $\Delta b = (1/15, 1, 1/3)$ and modified betas $b' = (25/6, -13/6, -20/3)$. The numerator magnitudes $|6b_2'| = 13$ and $|3b_3'| = 20$ are the integers that enter every cosmological formula. Every link is verified EXACT in Fraction arithmetic. The chain is Level 1 — it holds in any universe with the same gauge group. A structural finding: the cosmological integers 13 and 20 come from the SU(2) and SU(3) Dynkin coefficients $C_2 = 2/3$ and $C_3 = 1/3$, which do not contain the GUT normalization factor k_1. The cosmological program is independent of the GUT normalization convention.

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1. The Question

The beta unification program uses the integers 13, 19, 20, and 22 to predict the cosmological constant scale, the dark matter to baryon ratio, the Hubble constant, and the cosmic energy density fractions. These predictions hit measured values at sub-percent precision with zero cosmological input (PHYS-25, 47/47 checks).

Where do these integers come from? Can each be derived from the gauge group SU(3) × SU(2) × U(1) and its representations, with no measurement?

This paper traces each integer to its origin. The trace has seven links. Each link is verified EXACT in Python Fraction arithmetic — not to some number of digits, but exactly, with zero tolerance. The complete chain is Level 1. A civilization with different coupling constants but the same gauge group would derive the same integers.

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2. The SU(5) Embedding Condition

The Standard Model gauge group SU(3) × SU(2) × U(1) embeds into the grand unified group SU(5). The embedding requires a normalization condition: the U(1) hypercharge generator Y must be normalized to match the SU(2) isospin generator T_3. The condition is $\mathrm{Tr}(Y^2) = k_1 \times \mathrm{Tr}(T_3^2)$, where the traces run over one complete SM generation of 15 Weyl fermions in five representations.

The trace of Y^2 has five contributions. The left-handed quark doublet Q_L in (3,2,1/6) has 6 states each with Y = 1/6, contributing 6 × (1/36) = 1/6. The right-handed up quark u_R in (3,1,2/3) has 3 states at Y = 2/3, contributing 3 × (4/9) = 4/3. The right-handed down quark d_R in (3,1,−1/3) has 3 states at Y = −1/3, contributing 3 × (1/9) = 1/3. The left-handed lepton doublet L_L in (1,2,−1/2) has 2 states at Y = −1/2, contributing 2 × (1/4) = 1/2. The right-handed electron e_R in (1,1,−1) has 1 state at Y = −1, contributing 1. The total is 1/6 + 4/3 + 1/3 + 1/2 + 1 = 10/3.

The trace of T_3^2 is simpler. Only SU(2) doublets contribute. Q_L has 3 colors × 2 isospin states, each with $T_3 = \pm 1/2$, contributing 3 × 2 × (1/4) = 3/2. L_L has 2 states contributing 2 × (1/4) = 1/2. The total is 2.

The normalization factor: $k_1 = \mathrm{Tr}(T_3^2)/\mathrm{Tr}(Y^2)$ = 2/(10/3) = 6/10 = 3/5.

This is the root of the chain. The factor 3/5 enters the U(1) beta coefficient formula as a normalization constant. Its reciprocal 5/3 enters the GUT-normalized U(1) coupling: $\alpha_1 = (5/3) \times \alpha\_EM/\cos^2\theta\_W$.

(Backed by phys26_normalization.py S1: $\mathrm{Tr}(Y^2)$ = 10/3 EXACT, $\mathrm{Tr}(T_3^2)$ = 2 EXACT, k_1 = 3/5 EXACT.)

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3. The Dynkin Coefficients

From $k_1 = 3/5$, the one-loop beta shift coefficients for a vector-like fermion pair — a particle plus its antiparticle, both chiralities present — are determined by the gauge group structure and the Dynkin index normalization for Dirac fermions:

$C_1 = 2k_1/3 = 2(3/5)/3 = 2/5$ (for U(1), contains the GUT normalization)

$C_2 = 2/3$ (for SU(2), independent of k_1)

$C_3 = 1/3$ (for SU(3), independent of k_1)

The beta shift formulas for any representation (R_3, R_2, Y) under $SU(3) \times SU(2) \times U(1)$:

$\Delta b_1 = C_1 \times \dim(R_3) \times \dim(R_2) \times Y^2$

$\Delta b_2 = C_2 \times \dim(R_3) \times S_2(R_2)$

$\Delta b_3 = C_3 \times \dim(R_2) \times S_2(R_3)$

where S_2(fundamental of SU(N)) = 1/2 is the Dynkin index of the fundamental representation.

A structural observation that will matter in Section 8: C_2 and C_3 do not contain k_1. Only C_1 does. The SU(2) and SU(3) coefficients are determined by the Dynkin index normalization alone. The GUT normalization factor enters only the U(1) sector.

For the Cabibbo Doublet (3,2,1/6), the step-by-step computation of Δb_1: $(2/5) \times 3 = 6/5$, then $\times 2 = 12/5$, then $\times (1/6)^2 = (12/5) \times (1/36) = 12/180 = 1/15$. For Δb_2: $(2/3) \times 3 = 2$, then $\times (1/2) = 1$. For Δb_3: $(1/3) \times 2 = 2/3$, then $\times (1/2) = 1/3$.

Result: $\Delta b = (1/15, 1, 1/3)$. Matching the library values exactly.

(Backed by phys26_normalization.py S2: $\Delta b_1 = 1/15$ EXACT, $\Delta b_2 = 1$ EXACT, $\Delta b_3 = 1/3$ EXACT.)

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4. Three Independent Verifications

The coefficients (2/5, 2/3, 1/3) are verified by three independent tests, each from a different domain.

The MSSM gate: the same Dynkin formulas applied to the full MSSM particle content produce the MSSM beta coefficients (33/5, 1, −3), giving gap ratio $(33/5 - 1)/(1 - (-3)) = (28/5)/4 = 7/5 = 1.400$ exactly. This result has been verified by decades of independent computation in the GUT literature. If the coefficients were wrong, this gate would fail.

The Higgs cross-check: a single complex scalar uses coefficients exactly half the VL fermion values — (1/5, 1/3, 1/6). Applied to the SM Higgs (1,2,1/2): $\Delta b_1 = (1/5) \times 1 \times 2 \times (1/4) = 1/10$, $\Delta b_2 = (1/3) \times 1 \times (1/2) = 1/6$, $\Delta b_3 = (1/6) \times 2 \times 0 = 0$. These are the known Higgs contributions to the SM betas, verified in the democracy decomposition (PHYS-17, phys24_democracy.py 10/10). The relationship VL = 2 × complex scalar is exact for all groups and all representations.

Two routes to 1/15: Route A uses the direct Dynkin formula (2/5) $\times$ 3 $\times$ 2 $\times$ (1/36) = 1/15. Route B counts the VL pair as 2 complex scalars for U(1) purposes, giving 2 $\times$ (1/5) $\times$ 3 $\times$ 2 $\times$ (1/36) = 2 $\times$ (1/30) = 1/15. Both routes produce the same Fraction. The factor of 2 is the VL pair counting — Route A absorbs it into C_1 = 2/5, Route B makes it explicit.

(Backed by phys26_normalization.py S3: MSSM gap 7/5 EXACT. S4: Route A = 1/15 EXACT, Route B = 1/15 EXACT, Route A = Route B EXACT. S6: Higgs Δb_1 = 1/10 EXACT, Higgs Δb_2 = 1/6 EXACT.)

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5. The Modified Betas

Adding the Cabibbo Doublet shifts (1/15, 1, 1/3) to the SM betas (41/10, −19/6, −7):

b_1' = 41/10 + 1/15 = 123/30 + 2/30 = 125/30 = 25/6

b_2' = −19/6 + 6/6 = −13/6

b_3' = −21/3 + 1/3 = −20/3

The gap ratio: (25/6 − (−13/6)) / (−13/6 − (−20/3)) = (38/6) / (27/6) = 38/27 = 1.407. Distance from the measured gap ratio 1.358: 0.049.

(Backed by phys26_normalization.py S7: b_1' = 25/6 EXACT, b_2' = −13/6 EXACT, b_3' = −20/3 EXACT, gap = 38/27 EXACT.)

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6. The Integers

From the modified betas, the integers that enter the cosmological formulas:

|6 $\times$ b_2'| = |6 $\times$ (−13/6)| = 13. This is the numerator magnitude of the modified SU(2) beta coefficient. It appears in the dark matter ratio (22/13), the Hubble correction (20/13), the baryon density (2/(13 π)), the dark matter density (44/169 = 44/13^2), the weak mixing angle (3/13), and the cosmological constant exponent (39 = 3 $\times$ 13). |3 $\times$ b_3'| = |3 $\times$ (−20/3)| = 20. This is the numerator magnitude of the modified SU(3) beta coefficient. It appears in the Hubble correction (20/13). From the Yang-Mills gauge self-coupling — pre-Cabibbo-Doublet, from the SM structure:

2 $\times$ 11 = 22. The Yang-Mills integer 11 enters through b_2_gauge = −(11/3) $\times$ C_2(SU(2)) = −22/3. Twice this gives 22, which appears in the dark matter ratio (22/13) and the baryon density.

|6 $\times$ b_2 SM| = |6 $\times$ (−19/6)| = 19. The SM SU(2) beta numerator. It appears in the cosmological constant exponent (57 = 3 $\times$ 19) and the exact identity 57/39 = 19/13. N_gen = 3. The generation count from anomaly cancellation. It multiplies 19 and 13 to give the cosmological constant exponents 57 and 39.

Five primary integers: 11, 13, 19, 20, 3. Five derived integers: $22 = 2 \times 11$, $44 = 4 \times 11$, $169 = 13^2$, $57 = 3 \times 19$, $39 = 3 \times 13$. All ten traceable to the gauge group.

(Backed by phys26_normalization.py S7: $|6b_2'| = 13$ EXACT, $|3b_3'| = 20$ EXACT.)

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7. The Chain

The complete traceability chain from the SU(5) embedding to the cosmological integers:

Link 1: The SU(5) embedding condition requires $\mathrm{Tr}(Y^2) = k_1\,\mathrm{Tr}(T_3^2)$ over one generation.

Link 2: Explicit traces give $k_1 = \mathrm{Tr}(T_3^2)/\mathrm{Tr}(Y^2) = 2/(10/3) = 3/5$.

Link 3: The VL Dynkin coefficients follow: $C_1 = 2k_1/3 = 2/5$, $C_2 = 2/3$, $C_3 = 1/3$.

Link 4: Applied to (3,2,1/6): $\Delta b_1 = 1/15$, $\Delta b_2 = 1$, $\Delta b_3 = 1/3$.

Link 5: Modified SU(2) beta: $b_2' = -19/6 + 1 = -13/6$.

Link 6: Modified SU(3) beta: $b_3' = -21/3 + 1/3 = -20/3$.

Link 7: Extract integers: $|6b_2'| = 13$, $|3b_3'| = 20$.

Every link is verified EXACT. No measurement enters. The chain is Level 1.

The integers then enter the cosmological formulas. The cosmological constant scale is α^(3 × 19) or (α/3 π)^(3 × 13). The dark matter ratio is (22/13) π . The Hubble correction is α^2 π ^{2}(20/13). The baryon density is 2/(13 π). The dark matter density is 44/169. The weak mixing angle is approximately 3/13. Every formula traces back through this chain to the SU(5) embedding condition.

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8. The Structural Finding

The cosmological integers 13 and 20 enter through b_2' and b_3'. These modified betas use the Dynkin coefficients $C_2 = 2/3$ and $C_3 = 1/3$ to compute Δb_2 and Δb_3. Neither C_2 nor C_3 contains the GUT normalization factor $k_1 = 3/5$. The factor k_1 enters only $C_1 = 2/5$, which determines $\Delta b_1 = 1/15$.

This means the cosmological formulas are independent of the GUT normalization convention. If k_1 were different — say 2/5 instead of 3/5 — the U(1) coefficient would change from $C_1 = 2/5$ to $C_1 = 4/15$. This would change Δb_1 from 1/15 to 2/15, the gap ratio from 38/27 to 64/45, and the unification scale M_GUT. But b_2' would still be $-13/6$. And b_3' would still be $-20/3$. The integers 13 and 20 would be unchanged. Every cosmological formula would produce the same prediction.

The unification program (Track A) depends on all three beta shifts including the k_1-sensitive Δb_1. The cosmology program (Track B) depends only on Δb_2 and Δb_3, which are k_1-independent. The two tracks share the Cabibbo Doublet but are sensitive to different aspects of its representation theory.

This is a robustness finding. The cosmological predictions are insulated from the most convention-dependent part of the GUT calculation. They depend only on the SU(2) and SU(3) Dynkin indices of the (3,2,1/6) representation — quantities determined by $\dim(R_3)$, $\dim(R_2)$, $S_2(R_2)$, and $S_2(R_3)$, which have nothing to do with U(1) normalization.

(Backed by phys26_normalization.py S5: wrong coefficient 4/5 gives $\Delta b_1 = 2/15$ and gap 64/45 — different from 38/27 — but the integers 13 and 20 are unchanged.)

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9. What This Paper Does Not Claim

This paper derives the integers. It does not derive the cosmological formulas. The integers 13 and 20 are correctly traced to the gauge group. Whether they appear in cosmological formulas for physical reasons or by coincidence is the subject of PHYS-31 (statistical control). The chain documented here is Level 1 — it is about where the integers come from, not about where they go.

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10. What This Paper Seeds

The Dynkin formulas in Section 3 are the reference for every subsequent representation computation. PHYS-27 uses $b_1' = 25/6$ and $b_2' = -13/6$ for $\sin^2\theta$_W running. PHYS-28 uses the formulas to compute VL two-loop contributions. PHYS-38 uses the integers 13 and 20 for the loop expansion analysis. PHYS-40 tests whether GUT running produces $\sin^2\theta$_W = 3/13 = N_gen/$|b_2'$ numerator$|$. Every paper in the program that uses an integer from the Cabibbo Doublet cites this paper for the derivation.

The general formulas in Section 3 apply to any representation (R_3, R_2, Y). A future session testing a different BSM candidate — a VL lepton doublet, a scalar leptoquark, a color octet — uses these formulas with the appropriate quantum numbers. The Cabibbo Doublet is not the only representation that can be analyzed this way. It is the one that survives the elimination cascade.

PHYS-26: The Integer Traceability Chain. Seven links, all EXACT. 20/20 checks, zero failures. Published April 2, 2026. This paper is never edited after publication.

Appendix A: The Seven-Link Chain

Link	Input	Operation	Output	Script Check	Precision
1	SM generation (5 reps)	$Tr(Y^2)$ over 15 states	10/3	S1: $Tr(Y^2) = 10/3$	EXACT
1	SM generation (2 doublets)	$Tr(T_3^2)$ over doublets	2	S1: $Tr(T_3^2) = 2$	EXACT
2	$Tr(T_3^2)/Tr(Y^2)$	Division	$k_1 = 3/5$	S1: $k_1 = 3/5$	EXACT
3	$k_1 = 3/5$	$C_1 = 2k_1/3$	$C_1 = 2/5$	S3: MSSM gate 7/5	EXACT
4	C = (2/5, 2/3, 1/3), (3,2,1/6)	Dynkin formulas	$\Delta b = (1/15, 1, 1/3)$	S2: all three EXACT	EXACT
5	b_2 SM + Δb_2	−19/6 + 1	$b_2' = -13/6$	S7: $b_2' = -13/6$	EXACT
6	b_3 SM + Δb_3	−21/3 + 1/3	$b_3' = -20/3$	S7: $b_3' = -20/3$	EXACT
7	Extract numerators		$6b_2'$	,	$3b_3'$

Appendix B: The Trace Computation

Representation	Y	Y^2	States	$Tr(Y^2)$ contribution	T_3	$Tr(T_3^2)$ contribution
Q L (3,2,1/6)	1/6	1/36	6	1/6	$\pm$ 1/2 $\times$ 3 colors	3/2
u R (3,1,2/3)	2/3	4/9	3	4/3	0	0
d R (3,1,−1/3)	−1/3	1/9	3	1/3	0	0
L L (1,2,−1/2)	−1/2	1/4	2	1/2	$\pm$ 1/2	1/2
e R (1,1,−1)	−1	1	1	1	0	0
Total			**15**	**10/3**		**2**

$k_1 = 2/(10/3) = 6/10 = 3/5$.

Appendix C: Two Routes to 1/15

Step	Route A (Dynkin direct)	Route B (2 $\times$ scalar)
Starting coefficient	$C_1 = 2/5$	C_1 scalar = 1/5
$\times$ dim(R_3) = 3	6/5	3/5
$\times$ dim(R_2) = 2	12/5	6/5
$\times$ $Y^2 = 1/36$	12/180 = 1/15	6/180 = 1/30
$\times$ VL factor	(absorbed in C_1)	$\times$ 2 = 2/30 = 1/15
Result	**1/15**	**1/15**

Appendix D: Track Independence

Quantity	Depends on k_1?	Track A uses?	Track B uses?
$C_1 = 2/5$	**YES** (= $2k_1/3$)	Yes (Δb_1)	No
$C_2 = 2/3$	No	Yes (Δb_2)	Yes (integer 13)
$C_3 = 1/3$	No	Yes (Δb_3)	Yes (integer 20)
$\Delta b_1 = 1/15$	**YES**	Yes (gap ratio)	No
$\Delta b_2 = 1$	No	Yes (gap ratio)	Yes ($b_2' \to 13$)
$\Delta b_3 = 1/3$	No	Yes (gap ratio)	Yes ($b_3' \to 20$)
Gap ratio 38/27	**YES**	Yes (M GUT)	No
Integer 13	No	No	**Yes (6 formulas)**
Integer 20	No	No	**Yes (H_0 formula)**

Appendix E: General Formulas

For a vector-like fermion pair (R_3, R_2, Y):

Group	Formula	Coefficient
U(1)	$\Delta b_1 = (2/5) \times \dim(R_3) \times \dim(R_2) \times Y^2$	$2/5 = 2k_1/3$
SU(2)	$\Delta b_2 = (2/3) \times \dim(R_3) \times S_2(R_2)$	2/3
SU(3)	$\Delta b_3 = (1/3) \times \dim(R_2) \times S_2(R_3)$	1/3

For a single complex scalar (R_3, R_2, Y):

Group	Formula	Coefficient
U(1)	$\Delta b_1 = (1/5) \times \dim(R_3) \times \dim(R_2) \times Y^2$	$1/5 = k_1/3$
SU(2)	$\Delta b_2 = (1/3) \times \dim(R_3) \times S_2(R_2)$	1/3
SU(3)	$\Delta b_3 = (1/6) \times \dim(R_2) \times S_2(R_3)$	1/6

Relationship: VL fermion = 2 × complex scalar for all groups. The factor is exact and representation-independent.

Higgs cross-check: SM Higgs (1,2,1/2) as single complex scalar gives (1/10, 1/6, 0) — matching the known SM values.

Appendix F: Where Each Integer Goes

Integer	Value	Origin (this paper)	Cosmological formulas (PHYS-25)
13		$6b_2$'	
20		$3b_3$'	
22	2 × 11	Yang-Mills theorem	DM(22/13), Ω DM(44=2 × 22)
19		$6b_2$ SM	
3	N gen	Anomaly cancellation	Λ (3 × 19, 3 × 13), $\sin^2\theta$ W(3/13)

Appendix G: Verification Summary

Section	Checks	All EXACT?
S1: GUT normalization (k_1 = 3/5)	3	Yes
S2: Dynkin formulas (Δb = 1/15, 1, 1/3)	3	Yes
S3: MSSM gate (7/5)	1	Yes
S4: Two routes (both 1/15, A = B)	3	Yes
S5: Wrong convention (2/15, gap 64/45)	2	Yes

Section	Checks	All EXACT?
S6: Higgs cross-check (1/10, 1/6)	2	Yes
S7: Modified betas + integers (25/6, −13/6, −20/3, 38/27, 13, 20)	6	Yes
Total	**20**	**All 20 EXACT**

Supporting appendices A through G for PHYS-26. Every link in the chain is verified EXACT. Every integer traces to the gauge group. The cosmological program is independent of the GUT normalization convention.

Supporting Appendix Tables for PHYS-26

TABLE 26.1: THE SEVEN-LINK CHAIN — EXPANDED WITH INTERMEDIATE FRACTIONS

Link	Input	Step 1	Step 2	Step 3	Final Output	Check ID	Status
1a	Q L: 6 × (1/36)	u R: 3 × (4/9)	d R: 3 × (1/9)	L L: 2 × (1/4), e R: 1 × 1	$\mathrm{Tr}(Y^2)$ = 10/3	S1	EXACT
1b	Q L: 3 × 2 × (1/4)	L L: 2 × (1/4)	—	—	$\mathrm{Tr}(T_3^2)$ = 2	S1	EXACT
2	$\mathrm{Tr}(T_3^2)$ = 2	$\mathrm{Tr}(Y^2)$ = 10/3	2/(10/3) = 6/10	—	k_1 = 3/5	S1	EXACT
3a	k_1 = 3/5	2 × (3/5) = 6/5	(6/5)/3 = 6/15	= 2/5	C_1 = 2/5	S3 (via MSSM)	EXACT
3b	—	—	—	—	C_2 = 2/3	S2	EXACT
3c	—	—	—	—	C_3 = 1/3	S2	EXACT
4a	C_1=2/5, (3,2,1/6)	(2/5) × 3 = 6/5	× 2 = 12/5	× (1/36) = 12/180	Δb_1 = 1/15	S2	EXACT

Link	Input	Step 1	Step 2	Step 3	Final Output	Check ID	Status
4b	C_2=2/3, (3,2,1/6)	(2/3) × 3 = 2	× (1/2) = 1	—	Δb_2 = 1	S2	EXACT
4c	C_3=1/3, (3,2,1/6)	(1/3) × 2 = 2/3	× (1/2) = 1/3	—	Δb_3 = 1/3	S2	EXACT
5	b_2 SM = −19/6	+ Δb_2 = 6/6	−19/6 + 6/6	= −13/6	b_2' = −13/6	S7	EXACT
6	b_3 SM = −21/3	+ Δb_3 = 1/3	−21/3 + 1/3	= −20/3	b_3' = −20/3	S7	EXACT
7a	b_2' = −13/6		6 × (−13/6)		=	−13	
7b	b_3' = −20/3		3 × (−20/3)		=	−20	

14 sub-links. Every intermediate Fraction shown. Every output EXACT.

TABLE 26.2: THE FIVE SM REPRESENTATIONS — COMPLETE QUANTUM NUMBERS

Rep	SU(3)	SU(2)	Y	Q = T_3+Y	States	Color states	Name
Q L	3	2	1/6	(+2/3, −1/3)	6	3	Left-handed quark doublet
u R	3	1	2/3	+2/3	3	3	Right-handed up quark
d R	3	1	−1/3	−1/3	3	3	Right-handed down quark
L L	1	2	−1/2	(0, −1)	2	1	Left-handed lepton doublet
e R	1	1	−1	−1	1	1	Right-handed electron
Total					**15**		One complete generation

TABLE 26.3: Tr(Y^2) — EVERY TERM

Rep	Y	Y^2	States	Y^2 × States	Fraction	Decimal
Q L	1/6	1/36	6	6/36	1/6	0.16667
u R	2/3	4/9	3	12/9	4/3	1.33333
d R	−1/3	1/9	3	3/9	1/3	0.33333
L L	−1/2	1/4	2	2/4	1/2	0.50000
e R	−1	1	1	1	1	1.00000
Total			**15**		**10/3**	**3.33333**

TABLE 26.4: $Tr(T_3^2)$ — EVERY TERM

Rep	SU(2) rep	T_3 values	T_3^2 per state	Color mult.	States	Contribution
Q L	doublet	+1/2, −1/2	1/4	3	6	6 × 1/4 = 3/2
u R	singlet	0	0	3	3	0
d R	singlet	0	0	3	3	0
L L	doublet	+1/2, −1/2	1/4	1	2	2 × 1/4 = 1/2
e R	singlet	0	0	1	1	0
Total					**15**	**2**

TABLE 26.5: THE THREE DYNKIN COEFFICIENTS — DERIVATION CHAIN

Coefficient	Formula	Derivation	Contains k_1?	Used by Track
$C_1 = 2/5$	$2k_1/3$	2 × (3/5) / 3 = 6/15 = 2/5	**YES**	A only ($\Delta b_1 \rightarrow$ gap ratio)
$C_2 = 2/3$	2/3	Dynkin normalization for Dirac fermion, SU(2)	No	A and **B** ($\Delta b_2 \rightarrow 13$)
$C_3 = 1/3$	1/3	Dynkin normalization for Dirac fermion, SU(3)	No	A and **B** ($\Delta b_3 \rightarrow 20$)

The factor 2 in all three coefficients counts the VL pair (particle + antiparticle). The factor 1/3 (in C_2 and C_3) or $k_1/3$ (in C_1) is the Dynkin index normalization. The $k_1 = 3/5$ factor in C_1 is the GUT normalization converting Y to the SU(5)-normalized charge.

TABLE 26.6: THE CABIBBO DOUBLET DYNKIN COMPUTATION — EVERY STEP

Beta	Formula	$\times$ $\dim(R_3)=3$	$\times$ $\dim(R_2)=2$ or $S_2=1/2$	$\times$ $Y^2=1/36$	Result
Δb_1	$(2/5) \times 3 \times 2 \times (1/36)$	6/5	12/5	12/180 = 1/15	**1/15**
Δb_2	$(2/3) \times 3 \times (1/2)$	2	1	—	**1**
Δb_3	$(1/3) \times 2 \times (1/2)$	—	2/3	—	**1/3**

TABLE 26.7: THE MSSM GATE — COMPLETE

MSSM Parameter	Value	Source
b_1 MSSM	33/5 = 6.600	Standard MSSM literature
b_2 MSSM	1 = 1.000	Standard MSSM literature
b_3 MSSM	−3 = −3.000	Standard MSSM literature
$b_1 - b_2$	33/5 − 1 = 28/5	Exact Fraction
$b_2 - b_3$	1 − (−3) = 4	Exact Fraction
Gap ratio	(28/5)/4 = 28/20 = 7/5	EXACT
Known result	7/5 = 1.400	40 years of literature verification
Match	**EXACT**	Convention verified

TABLE 26.8: TWO ROUTES TO 1/15 — FRACTION ARITHMETIC

Step	Route A: Dynkin direct	Fraction	Route B: 2 × scalar	Fraction
1	Coefficient	2/5	Scalar coefficient	1/5
2	$\times \dim(R_3)$	$(2/5) \times 3 = 6/5$	$\times \dim(R_3)$	$(1/5) \times 3 = 3/5$
3	$\times \dim(R_2)$	$(6/5) \times 2 = 12/5$	$\times \dim(R_2)$	$(3/5) \times 2 = 6/5$
4	$\times Y^2$	$(12/5) \times (1/36) = 12/180$	$\times Y^2$	$(6/5) \times (1/36) = 6/180$
5	Simplify	12/180 = 1/15	Simplify	6/180 = 1/30
6	—	—	× VL factor 2	2/30 = 1/15
Result		**1/15**		**1/15**

Route A absorbs the factor of 2 into $C_1 = 2/5$. Route B makes the factor explicit. Same Fraction at every step where they overlap.

TABLE 26.9: THE WRONG CONVENTION — QUANTIFIED

Quantity	Correct ($C_1 = 2/5$)	Wrong ($C_1 = 4/5$)	Ratio wrong/correct
Δb_1	1/15 = 0.0667	2/15 = 0.1333	2 ×
b_1’	25/6 = 4.1667	127/30 = 4.2333	1.016 ×
b_1’ − b_2’	38/6 = 6.3333	192/30 = 6.4000	1.011 ×
b_2’ − b_3’	27/6 = 4.5000	27/6 = 4.5000	1.000 × (unchanged)
Gap ratio	38/27 = 1.4074	64/45 = 1.4222	1.011 ×
Distance from 1.358	0.049	0.064	1.30 × (30% worse)
Asymmetry $\Delta b_2/\Delta b_1$	15	15/2 = 7.5	0.50 × (halved)
Integer 13	13 (unchanged)	13 (unchanged)	1.00 ×
Integer 20	20 (unchanged)	20 (unchanged)	1.00 ×
DM/baryon prediction	(22/13) π = 5.317	(22/13) π = 5.317	1.00 ×
H_0 prediction	67.364	67.364	1.00 ×
Ω b prediction	0.04897	0.04897	1.00 ×

The wrong convention degrades unification (gap ratio 30% worse) but leaves cosmology completely unchanged. This is because the error is in C_1 (U(1)), and cosmological formulas use only C_2 (SU(2)) and C_3 (SU(3)) outputs.

TABLE 26.10: THE HIGGS CROSS-CHECK — SCALAR FORMULA VERIFICATION

Higgs (1,2,1/2)	Scalar formula	Computation	Result	Known SM value	Match
Δb_1	(1/5) × dim(R_3) × dim(R_2) × Y^2	(1/5) × 1 × 2 × (1/4)	2/20 = 1/10	1/10	EXACT
Δb_2	(1/3) × dim(R_3) × $S_2(R_2)$	(1/3) × 1 × (1/2)	1/6	1/6	EXACT

Higgs (1,2,1/2)	Scalar formula	Computation	Result	Known SM value	Match
Δb_3	$(1/6) \times \dim(R_2) \times S_2(R_3)$	$(1/6) \times 2 \times 0$	0	0	EXACT

Independent verification of the scalar counting convention. The Higgs is a known single complex scalar with known beta contributions from the democracy decomposition (PHYS-17). All three match.

TABLE 26.11: VL vs SCALAR — THE UNIVERSAL FACTOR OF 2

Group	VL coefficient	Scalar coefficient	Ratio	Representation-dependent?
U(1)	2/5	1/5	2	No
SU(2)	2/3	1/3	2	No
SU(3)	1/3	1/6	2	No

The factor of 2 is universal across all three gauge groups. It counts the VL pair: one particle plus one antiparticle. This is not a convention choice — it is the physics of having both chiralities present.

TABLE 26.12: MODIFIED BETAS — SM → SM + CD

Beta	SM value	+ Δb	Modified value	Exact Fraction	Decimal
b_1	41/10	+ 1/15	41/10 + 1/15 = 123/30 + 2/30	125/30 = 25/6	4.16667
b_2	−19/6	+ 1	−19/6 + 6/6	−13/6	−2.16667
b_3	−7 = −21/3	+ 1/3	−21/3 + 1/3	−20/3	−6.66667

Gap ratio: $(25/6 - (-13/6)) / (-13/6 - (-20/3)) = (25/6 + 13/6) / (-13/6 + 20/3) = (38/6) / (-13/6 + 40/6) = (38/6) / (27/6) = 38/27$.

TABLE 26.13: THE INTEGER EXTRACTION

Modified beta	Value	× clearing factor	Numerator magnitude	Integer	Cosmological appear-ances
b_2’	−13/6	× 6		−13	
b_3’	−20/3	× 3		−20	
b_2 SM	−19/6	× 6		−19	
b_2 gauge	−22/3	× 3/2		−11	× 2

TABLE 26.14: THE FIVE PRIMARY AND FIVE DERIVED INTEGERS

Type	Integer	Value	Origin	Chain length
Primary	11	Yang-Mills	$-(11/3)C_2(G)$, Lorentz + gauge + renorm.	1 link
Primary	13		b_2’ numerator	
Primary	19		b_2 SM numerator	
Primary	20		$3b_3$’	
Primary	3	N gen	SU(5) anomaly cancellation	1 link
Derived	22	2 × 11	2 × Yang-Mills	2 links
Derived	44	4 × 11	4 × Yang-Mills	2 links
Derived	169	13^2		b_2’ num
Derived	57	3 × 19	N gen ×	b_2 SM num
Derived	39	3 × 13	N gen ×	b_2’ num

TABLE 26.15: TRACK INDEPENDENCE — COMPLETE MATRIX

Quantity	Formula	Contains k_1?	Track A needs?	Track B needs?	If k_1 wrong, changes?
C_1	$2k_1/3$	**YES**	Yes	No	**YES**
C_2	2/3	No	Yes	Yes	No
C_3	1/3	No	Yes	Yes	No

Quantity	Formula	Contains k_1?	Track A needs?	Track B needs?	If k_1 wrong, changes?
Δb_1	$C_1 \times 3 \times 2 \times Y^2$	**YES**	Yes	No	**YES** → 2/15
Δb_2	$C_2 \times 3 \times S_2$	No	Yes	Yes	No
Δb_3	$C_3 \times 2 \times S_2$	No	Yes	Yes	No
b_1’	$41/10 + \Delta b_1$	**YES**	Yes	No	**YES** → 127/30
b_2’	$-19/6 + \Delta b_2$	No	Yes	**Yes (→ 13)**	No
b_3’	$-7 + \Delta b_3$	No	Yes	**Yes (→ 20)**	No
Gap ratio	$(b_1' - b_2')/(b_2' - b_3')$	**YES**	**Yes**	No	**YES** → 64/45
M GUT	From running	**YES**	**Yes**	No	**YES**
τ proton	$\propto$ M GUT^4	**YES**	**Yes**	No	**YES**
Integer 13		$6b_2$’		No	No
Integer 20		$3b_3$’		No	No
DM/baryon	(22/13) π	No	No	**Yes**	No
H_0 correction	$\alpha^2 \pi^2 (20/13)$	No	No	**Yes**	No
Ω b	2/(13 π)	No	No	**Yes**	No
Ω DM	44/169	No	No	**Yes**	No

18 quantities checked. Track A has 8 k_1-dependent quantities. Track B has 0 k_1-dependent quantities. The tracks are fully independent on the normalization convention.

TABLE 26.16: THE GENERAL FORMULAS — FOR ANY FUTURE REPRESENTATION

Vector-like fermion pair (R_3, R_2, Y):

Input needed	Symbol	Example: (3,2,1/6)	Example: (1,2,−1/2)	Example: (3,1,2/3)
dim(R_3)	d_3	3	1	3
dim(R_2)	d_2	2	2	1
$S_2(R_3)$	T_3	1/2	0	1/2
$S_2(R_2)$	T_2	1/2	1/2	0
Y	Y	1/6	−1/2	2/3
$\Delta b_1 = (2/5) \times d_3 \times d_2 \times Y^2$		**1/15**	**2/5** × (1/4) = **1/10**	(2/5) × 3 × (4/9) = **8/15**

Input needed	Symbol	Example: (3,2,1/6)	Example: (1,2,−1/2)	Example: (3,1,2/3)
$\Delta b_2 = (2/3) \times d_3 \times T_2$		**1**	$(2/3) \times (1/2)$ = **1/3**	**0**
$\Delta b_3 = (1/3) \times d_2 \times T_3$		**1/3**	**0**	$(1/3) \times (1/2)$ = **1/6**

Any future BSM candidate tested using these formulas with the appropriate quantum numbers. The Cabibbo Doublet (3,2,1/6) is the one that survives the elimination cascade. The others are shown for reference.

TABLE 26.17: VERIFICATION SUMMARY — BY SECTION WITH CHECK NAMES

#	Check Name	Expected	Got	Precision
1	S1: $k_1 = \mathrm{Tr}(T_3^2)/\mathrm{Tr}(Y^2) = 3/5$	3/5	3/5	EXACT
2	S1: $\mathrm{Tr}(Y^2)$ per generation = 10/3	10/3	10/3	EXACT
3	S1: $\mathrm{Tr}(T_3^2)$ per generation = 2	2	2	EXACT
4	S2: Δb_1 from Dynkin = library 1/15	1/15	1/15	EXACT
5	S2: Δb_2 from Dynkin = library 1	1	1	EXACT
6	S2: Δb_3 from Dynkin = library 1/3	1/3	1/3	EXACT
7	S3: MSSM gap ratio = 7/5	7/5	7/5	EXACT
8	S4: Route A = 1/15	1/15	1/15	EXACT
9	S4: Route B = 1/15	1/15	1/15	EXACT
10	S4: Route A = Route B	1/15	1/15	EXACT
11	S5: Wrong coefficient gives $\Delta b_1 = 2/15$	2/15	2/15	EXACT
12	S5: Wrong gap ≠ correct gap	64/45 ≠ 38/27	TRUE	EXACT
13	S6: Higgs $\Delta b_1 = 1/10$	1/10	1/10	EXACT
14	S6: Higgs $\Delta b_2 = 1/6$	1/6	1/6	EXACT
15	S7: $b_1' = 25/6$	25/6	25/6	EXACT
16	S7: $b_2' = -13/6$	−13/6	−13/6	EXACT
17	S7: $b_3' = -20/3$	−20/3	−20/3	EXACT
18	S7: gap = 38/27	38/27	38/27	EXACT
19	S7:	$6b_2'$	= 13	13
20	S7:	$3b_3'$	= 20	20

20/20 PASS. All EXACT. Zero numerical comparisons — every check is an exact Fraction match. This is the highest precision tier in the series: not "agrees to N digits" but "identical as rational numbers."

TABLE 26.18: WHAT CITES PHYS-26

Paper	What it uses from PHYS-26	Specific section
PHYS-27	b_1' = 25/6, b_2' = −13/6 for running	Section 5
PHYS-28	Dynkin formulas for VL two-loop	Section 3
PHYS-29	Gap ratio 38/27 for threshold parametrization	Section 5
PHYS-30	Modified betas for α s prediction	Section 5
PHYS-31	Integer 13 for statistical control pool	Section 6
PHYS-32	Integers 13, 22 for Set B Omega chain	Section 6
PHYS-33	Integers 19, 13 for Λ interpolation	Section 6
PHYS-34	Integers 20, 13 for per-transit mechanism	Section 6
PHYS-35	H_0 formula using 20/13	Section 6
PHYS-36	General formulas for representation testing	Section 3
PHYS-37	Dynkin indices for QCD correction analysis	Section 3
PHYS-38	Integers 13, 20 for loop expansion	Section 6, Section 7
PHYS-39	Integer 13 for remainder domain analysis	Section 6
PHYS-40	Integer 13 for $\sin^2\theta$ W = 3/13 test	Section 6

Every paper in the program cites PHYS-26. It is the reference for all Dynkin formulas and all integer traceability claims.

TABLE 26.19: CUMULATIVE VERIFICATION

Script	Session	Checks	Status	What it backs
phys26 normalization.py	4	**20/20**	**ALL EXACT**	**This paper**
phys25 platform.py	4	47/47	PASS	PHYS-25 map paper
beta unification test.py	4	15/15	PASS	Cosmological discovery
qed predicts gr.py	4	10/10	PASS	QED-to-GR scan 1
qed gr scan 2.py	4	10/10	PASS	QED-to-GR scan 2
phys24 lib.py self-test	4	21/21	PASS	Platform
phys24 lib test.py	4	148/148	PASS	DATA-4 verification
8 PHYS-24 demo scripts	4	62/62	PASS	PHYS-24 content
6 Session 3 scripts	3	98/98	PASS	Session 3 ground

Script	Session	Checks	Status	What it backs
Grand total		**431/431**	**ZERO FAILURES**	**Complete series**

End of supporting appendix tables for PHYS-26. 19 tables. Every link in the seven-link chain is documented with intermediate Fractions. Every check is listed individually. The track independence matrix shows 18 quantities with their k_1 dependence. The general formulas are published for any future representation. Every paper in the program (PHYS-27 through PHYS-40) cites this paper. Grand total: 431/431, zero failures, 20/20 in this paper ALL EXACT.

[**HOWL-PHYS-26-2026**] The Integer Traceability Chain. Github: https://github.com/ghowland/papers/tree/main/papers/P
PHYS-26-2026

[**HOWL-PHYS-1-2026**] The Inertial Vortex. Github: https://github.com/ghowland/papers/tree/main/papers/PHYS/HOWL
PHYS-1-2026

[**HOWL-PHYS-13-2026**] Gauge Coupling Unification and Minimal BSM Content. Github: https://github.com/ghowland/papers/tree/main/papers/PHYS/HOWL-PHYS-13-2026

[**HOWL-PHYS-21-2026**] The Level 1 / Level 2 Boundary — From Observed Structure to Unobserved Particles. Github: https://github.com/ghowland/papers/tree/main/papers/PHYS/HOWL-PHYS-21-2026

[**HOWL-PHYS-24-2026**] The Session 3 Operational Lexicon. Github: https://github.com/ghowland/papers/tree/main/pape
PHYS-24-2026

[**HOWL-PHYS-25-2026**] The Session 4 Operational Map — From Gauge Integers to Cosmological Predictions. Github: https://github.com/ghowland/papers/tree/main/papers/PHYS/HOWL-PHYS-25-2026